

MONTAUBAN MORIN-VERDRINES AP, France

WMO: 075400

Lat: 44.0281N

Lon: 1.3767E

Elev: 107

StdP: 100.05

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.6	(c) -2.9	(d) -8.2	(e) 1.9	(f) -0.7	(g) -6.0	(h) 2.3	(i) -0.4	(j) 10.3	(k) 10.6	(l) 8.3	(m) 9.6	(n) 1.2	(o) 280	(p) 0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.6	(c) 34.4	(d) 21.3	(e) 32.2	(f) 20.8	(g) 30.5	(h) 20.1	(i) 22.7	(j) 30.8	(k) 21.8	(l) 29.8	(m) 21.0	(n) 28.5	(o) 2.9	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	14.9	26.3	19.2	14.2	25.1	18.5	13.5	24.2	67.2	30.8	64.1	30.0	61.3	28.7	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	6.8	5.7	-6.6	37.9	2.2	2.1	-8.2	39.4	-9.5	40.6	-10.7	41.8	-12.3	43.4
			-7.0	24.2	2.2	0.9	-8.6	24.8	-9.9	25.4	-11.1	25.9	-12.7	26.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	5.5	6.3	9.9	13.0	16.2	20.7	22.6	22.2	18.9	15.0	9.4	6.0		
	DBStd	6.99	3.60	3.71	3.24	2.95	3.06	3.39	2.92	3.07	3.09	3.73	3.70	3.83		
	HDD10.0	494	143	110	42	7	1	0	0	0	0	7	54	131		
	HDD18.3	2050	397	338	262	162	81	14	2	3	29	112	268	383		
	CDD10.0	1898	4	6	38	96	193	321	390	377	268	162	37	7		
	CDD18.3	413	0	0	0	1	14	86	134	122	47	9	0	0		
	CDH23.3	4193	0	0	2	43	179	859	1342	1212	483	73	0	0		
	CDH26.7	1644	0	0	0	6	38	332	576	533	151	9	0	0		
(14) Wind	WSAvg	2.3	2.2	2.5	2.8	2.7	2.7	2.4	2.4	2.2	1.9	2.0	2.1	1.9		
(15) Precipitation	PrecAvg	743	71	49	56	80	78	59	50	59	53	55	73	60		
	PrecMax	999	134	106	125	161	156	135	131	146	136	98	150	117		
	PrecMin	555	8	8	21	12	21	17	11	13	18	14	16	11		
	PrecStd	108	39	29	28	44	36	30	28	31	29	24	34	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	18.6	23.5	27.5	29.8	36.0	36.4	38.1	32.6	27.9	21.1	16.6		
		MCWB	12.5	11.8	13.2	16.6	18.6	21.7	21.6	21.6	20.0	18.4	15.4	12.6		
	2%	DB	13.8	16.1	20.4	24.4	27.6	32.4	34.0	34.6	30.4	25.3	19.0	14.6		
		MCWB	11.4	10.8	12.6	15.1	18.1	21.2	21.4	21.2	19.1	17.9	14.5	11.6		
	5%	DB	12.4	14.2	18.2	22.1	25.2	30.2	31.9	32.1	28.3	23.4	17.1	13.0		
		MCWB	10.6	10.0	11.6	14.3	17.3	20.2	20.8	20.8	18.2	17.1	13.5	11.0		
	10%	DB	11.0	12.4	16.2	19.8	23.0	28.3	30.1	29.7	26.2	21.6	15.2	11.7		
		MCWB	9.5	9.3	10.8	13.1	16.0	19.3	20.2	20.1	17.8	16.6	12.7	10.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	12.6	14.6	17.4	20.6	23.6	23.6	23.6	21.8	20.0	16.2	13.4		
		MCDB	14.4	16.0	20.0	25.1	27.0	31.8	32.8	32.5	29.9	25.6	19.9	15.3		
	2%	WB	12.1	11.5	13.4	16.0	19.0	22.3	22.6	22.3	20.5	18.9	15.0	12.2		
		MCDB	13.4	14.9	18.6	22.7	25.6	30.3	30.8	30.7	27.6	23.1	17.9	14.0		
	5%	WB	10.9	10.6	12.5	14.8	17.9	21.3	21.8	21.5	19.5	18.0	13.9	11.2		
		MCDB	12.1	13.4	17.0	20.5	23.7	28.6	29.5	29.7	25.9	21.7	16.3	12.8		
	10%	WB	9.7	9.6	11.5	13.7	16.7	20.2	20.9	20.6	18.6	17.1	13.0	10.2		
		MCDB	10.8	12.1	15.3	18.6	21.7	26.6	27.9	28.3	24.3	21.0	15.2	11.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	9.3	10.5	10.7	10.7	11.3	11.6	12.0	12.2	10.9	8.0	7.5		
		MCDBR	9.0	13.0	15.4	15.7	15.2	15.2	15.9	17.0	16.0	13.6	11.1	9.7		
	5% WB	MCWBR	6.7	8.2	8.6	7.8	7.2	6.1	5.7	5.9	6.3	7.0	7.1	7.1		
		MCWBR	7.3	11.1	12.7	13.5	12.9	13.7	13.6	13.8	13.0	11.1	9.5	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.334	0.371	0.400	0.406	0.418	0.410	0.409	0.378	0.365	0.338	0.318		
		taud	2.478	2.419	2.325	2.245	2.282	2.286	2.308	2.321	2.389	2.435	2.462	2.482		
	Ebn at Noon		806	855	863	863	866	855	857	844	845	807	772	769		
		Edh at Noon	71	89	111	131	130	130	126	120	104	87	72	65		
(44) All-Sky Solar Radiation	RadAvg		1.40	2.35	3.55	4.63	5.36	6.18	6.12	5.43	4.34	2.85	1.65	1.30		
	RadStd		0.15	0.27	0.36	0.53	0.40	0.48	0.44	0.34	0.25	0.23	0.19	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page